

ip-eval Race Results

candidate-expansion tranche · 2026-07-28 · racetrack/ip-eval · latest run per stage
IP-incumbent rows are highlighted ★ — every score comes from a real race on self-authored fixtures with gold oracles.

14

STAGES

58

RACE RUNS ON
DISK

61

CANDIDATES
RACED

29

HONEST-
UNAVAILABLE

5

DEFECTS
FIXED TODAY

Stage leaderboards

Extract

run3 · 3/6 gate PASS

S1 extraction: 2 docs x 5 formats + adversarial pack (gate: mean_f1>=0.3, >=2 adversarial passes)

pandoc-pypandoc

PASS

0.7989

mean F1 0.799

heading ret. 0.800

p50 ms 0

★ ip-incumbent

PASS

0.738

mean F1 0.738

heading ret. 0.800

p50 ms 44

naive-baseline

PASS

0.5875

mean F1 0.588

heading ret. 0.600

p50 ms 0

docx2python

FAIL

0

mean F1 0.200

heading ret. 0.200

p50 ms 0

mineru

FAIL

0

mean F1 0.000

heading ret. 0.000

p50 ms 0

undoc

FAIL

0

mean F1 0.200

heading ret. 0.200

p50 ms 0

Split

run4 · 2/4 gate PASS

S2 section splitting vs gold headings (gate: mean_f1>=0.3)

★ ip-incumbent

PASS

0.7778

mean F1 0.778

gamma F1 0.333

naive-baseline

PASS

0.6275

mean F1 0.627

gamma F1 0.000

haystack-splitter

FAIL

0

mean F1 0.000

gamma F1 0.000

paperspine-split

FAIL

0

mean F1 0.000

gamma F1 0.000

► honest-unavailable (3)

Concepts

run6 · 5/5 gate PASS

S3 concept extraction/ontology/nested suppression on incumbent-split sections (gate: f1>0)

★ ip-incumbent

PASS

0.7053

mean F1 0.571

type acc 1.000

nested 0.750

nemo-gliner

PASS

0.4896

mean F1 0.472

type acc 0.375

nested 0.750

naive-baseline

PASS

0.3791

mean F1 0.444

type acc 0.000

nested 0.750

spacy-nounchunks

PASS

0.1799

mean F1 0.112

type acc 0.000

nested 0.750

presidio

PASS

0.04

mean F1 0.067

type acc 0.000

nested 0.000

► honest-unavailable (4)

Tablify

run2 · 6/6 gate PASS

S3b tabilification: documents/sections/concepts/research tables + metric records + link integrity + UX-link probe (gate: zero fabricated metrics, links >= 0.5)

★ ip-incumbent

PASS

0.5614

docs F1 1.000

concepts F1 0.571

metrics F1 0.000

genie-worksheets

PASS

0.4773

docs F1 1.000

concepts F1 0.000

metrics F1 0.565

md-table-parser

PASS

0.4773

docs F1 1.000

concepts F1 0.000

metrics F1 0.565

openscience-eval

PASS

0.4773

docs F1 1.000

concepts F1 0.000

metrics F1 0.565

sheetjs-xlsx

PASS

0.4773

docs F1 1.000

concepts F1 0.000

metrics F1 0.565

naive-baseline

PASS

0.3518

docs F1 1.000

concepts F1 0.000

metrics F1 0.000

► honest-unavailable (1)

Retrieval

run6 · 10/10 gate PASS

S4 local-vault sentence retrieval vs gold evidence (gate: recall@5>0)

annoy

PASS

0.3564

recall@5 0.356 hard recall@5 1.000 p50 ms 284

haystack-bm25

PASS

0.3564

recall@5 0.356 hard recall@5 1.000 p50 ms 892

★ ip-incumbent

PASS

0.3564

recall@5 0.356 hard recall@5 0.833 p50 ms 88

librer

PASS

0.3564

recall@5 0.356 hard recall@5 0.833 p50 ms 2

ragbits-rrf

PASS

0.3564

recall@5 0.356 hard recall@5 0.833 p50 ms 0

rank-bm25

PASS

0.3564

recall@5 0.356 hard recall@5 1.000 p50 ms 211

tantivy-py

PASS

0.3564

recall@5 0.356 hard recall@5 1.000 p50 ms 101

faiss-hashdense

PASS

0.345

recall@5 0.345 hard recall@5 0.667 p50 ms 429

naive-baseline

PASS

0.3362

recall@5 0.336 hard recall@5 0.833 p50 ms 0

aann

PASS

0.1646

recall@5 0.165 hard recall@5 0.833 p50 ms 817

► honest-unavailable (7)

Graph

run4 · 5/5 gate PASS

S5 concept graph vs gold co-occurrence edges (gate: node_f1>0)

graphify-graph

PASS

0.9833

node F1 1.000 edge prec 0.944 edge recall 1.000

naive-baseline

PASS

0.9833

node F1 1.000 edge prec 0.944 edge recall 1.000

networkx-cooc

PASS

0.9833

node F1 1.000 edge prec 0.944 edge recall 1.000

★ ip-incumbent

PASS

0.9714

node F1 1.000 edge prec 0.905 edge recall 1.000

jaccard-graph

PASS

0.9556

node F1 1.000 edge prec 0.852 edge recall 1.000

► honest-unavailable (3)

Drafting

run5 · 5/5 gate PASS

S6 grounded drafting:
coverage/binding/fidelity/completeness (gate: zero
fabricated metrics)

paperspine-draft

PASS

0.9325

draft score 0.932 citation bind 0.775

metric fidelity 1.000

★ ip-incumbent

PASS

0.9077

draft score 0.908 citation bind 0.692

metric fidelity 1.000

sumy-textrank-citepost

PASS

0.8328

draft score 0.833 citation bind 0.443

metric fidelity 1.000

naive-baseline

PASS

0.7

draft score 0.700 citation bind 0.000

metric fidelity 1.000

sumy-lsa-citepost

PASS

0.7

draft score 0.700 citation bind 0.000

metric fidelity 1.000

► honest-unavailable (2)

Dedup

run6 · 1/4 gate PASS

S7 v2: required merges (punct variant) AND forbidden
merges (number/entity swaps) (gate: unique+swaps kept)

naive-baseline

PASS

0.6667

dedup score 0.667 punct merged 0 swaps kept 1

graphify-dedup

FAIL

0

dedup score 0.000 punct merged 0 swaps kept 0

★ ip-incumbent

FAIL

0

dedup score 0.667 punct merged 1 swaps kept 0

last30days-dedupe

FAIL

0

dedup score 0.667 punct merged 1 swaps kept 0

Export Docx

run4 · 4/4 gate PASS

S8 v2 DOCX: validity 0.35 + footnotes 0.15 + styles 0.10 +
heading styles 0.15 + text fidelity 0.25 (gate: valid)

★ ip-incumbent

PASS

1

docx score 1.000 text fidelity 1.000 footnotes 2

pandoc-docx

PASS

0.9937

docx score 0.994 text fidelity 0.975 footnotes 2

python-docx

PASS

0.8458

docx score 0.846 text fidelity 0.983 footnotes 0

docxtpl-minimal

PASS

0.8436

docx score 0.844 text fidelity 0.974 footnotes 0

► honest-unavailable (3)

Export Bibtex

run5 · 1/1 gate PASS

S8 v2 BibTeX: gold coverage + adversarial refs
(unicode/braces/missing year) (gate: emits entries)

★ ip-incumbent

PASS

1

coverage 1.000 entries 4 hard score 1.000

► honest-unavailable (1)

Persistence

run7 · 5/6 gate PASS

P10: torn-read atomicity (gate), lost updates, corrupt-file recovery, write p50

★ ip-incumbent

PASS

0.85

write p50 ms 2.34 torn reads 0 lost updates 0

opendal-store

PASS

0.85

write p50 ms 0.89 torn reads 0 lost updates 0

sqlite-store

PASS

0.85

write p50 ms 0.07 torn reads 0 lost updates 0

pouchdb-store

PASS

0.775

write p50 ms 0.97 torn reads 0 lost updates 0

turso-store

PASS

0.5

write p50 ms 0.39 torn reads 0 lost updates 20

naive-baseline

FAIL

0

write p50 ms 0.03 torn reads 376 lost updates 21

► honest-unavailable (2)

Api

run3 · 2/3 gate PASS

A10: live HTTP probes — 404s, malformed JSON 4xx, type errors, 60MB oversize, CORS, error leakage (gate: malformed => 4xx)

★ ip-incumbent

PASS

0.625

HTTP probes 62% p50 ms 11

hayhooks-api

PASS

0.5714

HTTP probes 57% p50 ms 18

fastapi-naive

FAIL

0

HTTP probes 71% p50 ms 5

Security

run2 · 1/1 gate PASS

S10: path-traversal confinement, planted prompt injection not obeyed, npm audit criticals (gate: traversal+injection)

★ ip-incumbent

PASS

1

checks 100% npm high 1

► honest-unavailable (3)

Frontend

run1 · 1/1 gate PASS

F11: playwright probes — routes, console, keyboard, responsive overflow, deep-link refresh (gate: routes render)

ip-frontend-playwright

PASS

1

UI probes 100%

Defects fixed in this tranche (before → after)

CANDIDATE	STAGE	WAS	NOW
annoy	retrieval	missing numpy in candidate PIP list	runs clean: recall@5 0.3564, ties the 7-way lead (run6)
turso-store	persistence	`nonlocal` SyntaxError + libsql lock retry	runs: gate PASS, score 0.5 — 20 lost updates under concurrency exposed (run7)
opendal-store	persistence	`nonlocal` SyntaxError	runs: score 0.85, ties ip-incumbent / sqlite at the top (run7)

hayhooks-api	api	server never came up on :8973; stderr was DEVNULL'd	server starts via hayhooks_cli with captured log: gate PASS, 0.5714 (run3)
nemo-gliner	concepts	TimeoutExpired escaped available(); crashed the stage (run5)	HF_HUB_OFFLINE fast path + exception-proof available(): gate PASS, 0.4896 (run6)

End-to-end pipeline (run2)

chain: extract(md) -> split -> concepts -> research -> combine -> docx+bibtex · wall 0.58s

s1_extract_f1 1

s2_boundary_f1 1

s3_concept_f1 0.5714

s3_type_accuracy 1

s4_recall_at_5 0.292

s7_unique_preserved 1

s8_docx_valid 1

s8_bibtex_coverage 1

Session notes

- Mid-session the whole ip-eval/ tree was accidentally deleted by an external session and fully restored; venvs re-provisioned on demand.
- Retrieval grew from a 5-way to a **7-way tie at 0.3564** (annoy and haystack-bm25 joined IP, librer, ragbits-rrf, rank-bm25, tantivy-py) once the haystack venv existed.
- Drafting: paperspine-draft 0.9325 took the crown from IP (0.9077). Graph: graphify-graph/naive/networkx 0.9833 > IP 0.9714.
- Persistence: IP = sqlite = opendal 0.85 > pouchdb 0.775 > turso 0.5 > naive FAIL (376 torn reads, 21 lost updates).
- race extract deliberately not re-run: new extract candidates are all honest-unavailable and the frontier is unchanged (OCR image lane, docker lane, R10 SLO/fault rows).